

Unit	Abbreviation	Description
Metre	m	Approximately three feet or one yard
Centimetre	cm	1/100 of a metre, about half of an inch
Millimetre	mm	1/1,000 of a metre
Micrometre	μm	1/1,000,000 of a metre, often called a micron — most integrated circuits are at this scale
Nanometre	nm	1/1,000,000,000 of a metre, the size of a single molecule

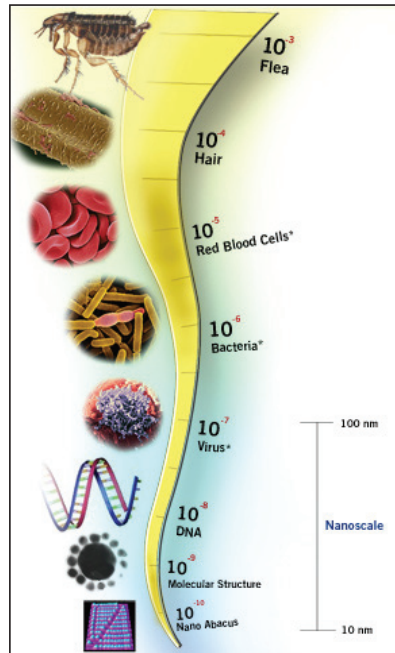
Scale of things

To actually get a feel of how small “nano” is, let’s put things in perspective here. As stated previously, a nanometre is one-billionth of a metre. Everyone struggles to imagine this very small scale, but you can get an idea through comparison.

- ▶ A sheet of paper is about 100,000 nms thick
- ▶ A strand of human DNA is 2.5 nms in diameter
- ▶ There are 25,400,000 nms in 1 inch
- ▶ A human hair is approximately 80,000nms wide
- ▶ A single gold atom is about a third of a nanometre in diameter
- ▶ On a comparative scale, if the diameter of a marble was 1nm, then the diameter of the Earth would be about 1m
- ▶ One nanometre is about as long as your fingernail grows in one second

For better understanding, we’ll look at nanotechnology in three parts:

- ▶ Seeing at the nanoscale
- ▶ Working at the nanoscale
- ▶ Manufacturing at the nanoscale



Scale of the things we're talking about